6. **Word2Vec** → Feature Representation. → Google

Word2vec is a technique for natural language processing published in 2013. The word2vec algorithm uses a neural network model to learn word associations from a large corpus of text. Once trained, such a model can detect synonymous words to suggest additional words for a partial sentence. As the name implies, word2vec represents each distinct word with a particular list of numbers called a vector.

Vocabulary → unique words → Corpus.

	Boy	Girl	King	Queen	Apple	Mango
Gender	-1	1	0.92	+0.93	0.01	0.05
Royal	0.01	0.02	0.95	0.96	0.01	0.02
Age	0.03	0.02	0.75	0.68	0.95	0.96
Food	-	-	-	-	0.91	0.92
⋮	-	-	-	-	-	-
nth	-	-	-	-	-	-

Feature Representation

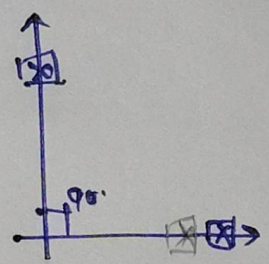
300 dimensions

[King - Boy + Queen] = Girl

King [0.95, 0.96] Man [0.95 0.98]

Queen [-0.96, 0.95] Woman [-0.94, -0.96]

KING - MAN + QUEEN = WOMEN.

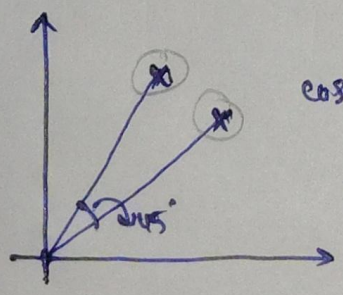


Distance = 1 - 0
= 1

Distance = 1 - cos90°
= 1 - 1 = 0

IronMan Avengers.

Cosine Similarity



Kinda similar.

Distance = 1 - cosine similarity

cosine simi = cos45° = $\frac{1}{\sqrt{2}}$

= 0.7071

Distance = 1 - 0.7071

= 0.29

[Action, Comic, Comedy]

